3D brain plotting with ggseg3d :: Cheatsheet

Interactive brain surfaces as an htmlwidget

Plotting

A brain in three dimensions

ggseg3d renders an atlas onto a brain mesh and returns an **htmlwidget** — draggable in RStudio, a Quarto document or a Shiny app.

```
library(ggseg3d)

ggseg3d(atlas = dk()) |>
  pan_camera("left lateral")
```

```
ggseg3d(
  .data = NULL, atlas = dk(),
  label_by = "region", text_by = NULL,
  colour_by = "colour", palette = NULL,
  na_colour = "darkgrey", na_alpha = 1, ...
)
```

- Atlases are **functions**: `dk()`, not `dk`.
- The result is a widget. Every option below is a **pipe step**, not an argument.
- `label_by` sets the hover label, `text_by` adds a second hover line.

Your data on the surface

Pass a data frame with a `region` column and name the column to colour by.

```
scores <- data.frame(
  region = c("precentral", "insula"),
  score = c(1.2, -0.4)
)

ggseg3d(
  .data = scores,
  atlas = dk(),
  colour_by = "score",
  palette = c("#f2f0ef", "#5e3c58")
) |>
  pan_camera("left lateral")
```

- `palette` takes colour names or hex codes, interpolated across the range.
- A **named numeric** palette pins colours to values: `c("#f2f0ef" = 0, "#5e3c58" = 3)`.
- Regions missing from `.data` take `na_colour`; `na_alpha` fades them.

How each atlas type is drawn

Type	Mesh	Colouring
cortical	one shared mesh per hemisphere	per vertex
cerebellar	one shared SUIT mesh	per vertex
subcortical	one mesh per structure	per face
tract	tube meshes built at render time	by orientation

Tract atlases encode direction as colour: red left–right, green anterior–posterior, blue superior–inferior.

Positioning

Cameras

```
ggseg3d(atlas = dk()) |>
  pan_camera("right medial")

# or an explicit x, y, z position
ggseg3d(atlas = dk()) |>
  pan_camera(c(-350, 0, 0))
```

Presets are `"<hemisphere> <view>"`, with an underscore accepted in place of the space:

	lateral	medial	superior	inferior	anterior	posterior
left	✓	✓	✓	✓	✓	✓
right	✓	✓	✓	✓	✓	✓

Which hemispheres and surfaces

```
ggseg3d(atlas = dk(), hemisphere = "left")

ggseg3d(atlas = dk(), surface = "pial")
```

`"inflated"` is the default and ships with **ggseg.formats**. The rest come from **ggseg.meshes**, an optional install:

```
install.packages(
  "ggseg.meshes",
  repos = "https://ggsegverse.r-universe.dev"
)

ggseg.meshes::available_cortical_surfaces()
#> pial, white, midthickness, semi-inflated,
#> sphere, smoothwm, orig

ggseg.meshes::available_cerebellar_surfaces()
#> suit_flat
```

`resolve_brain_mesh(hemisphere, surface, brain_meshes)` is the single entry point — pass `brain_meshes` to render on a mesh of your own.

A glass brain around a subcortical atlas

```
ggseg3d(atlas = aseg()) |>
  add_glassbrain("left", opacity = 0.2) |>
  pan_camera("left lateral")
```

```
add_glassbrain(
  p, hemisphere = c("left", "right"),
  surface = "inflated", colour = "#CCCCCC",
  opacity = 0.3, brain_meshes = NULL
)
```

Appearance

Widget options

Every one of these takes the widget and returns it, so they chain.

```
ggseg3d(atlas = dk()) |>
  set_background("#1a2a2e") |>
  set_edges(colour = "white", width = 1) |>
  set_legend(show = FALSE) |>
  set_dimensions(width = 800, height = 600) |>
  set_flat_shading(flat = TRUE) |>
  set_orthographic(ortho = TRUE, frustum_size = 220) |>
  set_positioning("centered") |>
  pan_camera("left lateral")
```

Step	Does
`set_background()`	canvas colour
`set_edges()`	region boundary lines
`set_legend()`	show or hide the colour legend
`set_dimensions()`	widget size in pixels
`set_flat_shading()`	faceted rather than smooth shading
`set_orthographic()`	drop perspective
`set_positioning()`	`"anatomical"` or `"centered"`

Saving a still image

```
ggseg3d(atlas = dk()) |>
  pan_camera("left lateral") |>
  snapshot_brain(
    "brain.png",
    width = 600, height = 500, zoom = 2
  )
```


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`snapshot_brain()` drives a headless browser, so it needs **webshot2** and Chrome. For a plot destined for print, `ggsegray()` renders the same scene through **rgl** instead:

```
ggsegray(atlas = dk(), colour_by = "colour") |>
  pan_camera("right lateral")
```

Shiny

Widget bindings

```
ui <- fluidPage(
  ggseg3dOutput(
    "brain",
    width = "100%",
    height = "600px"
  )
)

server <- function(input, output, session) {
  output$brain <- renderGgseg3d({
    ggseg3d(
      .data = scores(),
      atlas = dk(),
      colour_by = "score"
    )
  })
}
```

Other atlases

Every atlas on the r-universe works the same way, so long as it carries 3D data. `plot(atlas)` and the atlas's own print method say whether it does.

```
install_ggseg_atlas("ggsegYeo2011")

library(ggsegYeo2011)
ggseg3d(atlas = yeo7()) |>
  pan_camera("left lateral")
```

A cortical atlas built only from 2D polygons has no vertices, and ggseg3d errors rather than guessing at them.

Cerebellum and tracts

```
# cerebellum, on the shared SUIT mesh
ggseg3d(atlas = suit())

# tracts, coloured by orientation
ggseg3d(atlas = tracula()) |>
  add_glassbrain(opacity = 0.1)
```

Updating without a redraw

Rebuilding the widget reloads the whole mesh. These proxies change one thing in place:

```
updateGgseg3dCamera(session, "brain", "right lateral")
updateGgseg3dBackground(session, "brain", "#1a2a2e")
```

Gotchas

- `dk` is a function; `dk()` is the atlas.
- `label`, `text` and `colour` were renamed to `label_by`, `text_by` and `colour_by` in 2.1.0. The old names still work, with a warning.
- A widget cannot be printed to PDF. Use `snapshot_brain()` or `ggsegray()` for static output.
- Cortical meshes share vertex indices across regions and so cannot be decimated; subcortical meshes can.
- Asking for a non-`"inflated"` surface without **ggseg.meshes** installed is an error, not a silent fallback.

In documents

A widget needs a JavaScript-capable output. In Quarto and R Markdown it works in `html` formats, and in `pdf` only through a static snapshot:

````
```{r}
#| eval: false
ggseg3d(atlas = dk())
```
````

Guard it when a document renders to both:

```
#| eval: !expr knitr::is_html_output()
```

Each widget carries its own copy of the mesh, so a page with many brains gets heavy quickly — prefer one widget the reader can rotate over a grid of static views.

Colour and legend

`colour_by` names the column to map; `palette` says how.

```
# interpolated across the data range
palette = c("#f2f0ef", "#a8c5cb", "#5e3c58")

# pinned to specific values
palette = c(
  "#f2f0ef" = 0,
  "#a8c5cb" = 1.5,
  "#5e3c58" = 3
)
```

Leave `colour_by = "colour"` — the default — to draw the atlas in its own palette instead. `set_legend(FALSE)` drops the colour bar when the figure has its own caption.

Hover text

```
ggseg3d(
  .data = scores,
  atlas = dk(),
  colour_by = "score",
  label_by = "region",
  text_by = "score"
)
```

`label_by` is the bold first line of the tooltip, `text_by` a second line below it. Both name columns of `.data` or of the atlas `core`.

2D or 3D?

Want	Reach for
a figure for a paper	**ggseg**
a brain readers can rotate	**ggseg3d**
cortical surface anatomy	**ggseg3d**, inflated or pial
subcortical slices	**ggseg**, `aseg()`
subcortical volumes in space	**ggseg3d**, `aseg()`
tract orientation	**ggseg3d**, `tracula()`

When nothing appears

- **Blank panel in RStudio.** The widget needs the Viewer pane; try `print()` in a browser with `htmlwidgets::saveWidget()`.
- **Everything grey.** No region in `.data` matched the atlas. Compare `unique(.data$region)` with `atlas_regions(atlas)`.
- **An error about vertices.** The atlas has 2D geometry only. Use **ggseg**, or rebuild the atlas with **ggseg.extra**.
- **Slow first draw.** The mesh is loaded once per session; later widgets on the same surface are quicker.

Learn more

- `https://ggsegverse.github.io` — ecosystem documentation
- `vignette("ggseg3d")` — longer worked examples
- `https://github.com/ggsegverse/ggseg3d` — source and issues

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